

Agent Memory Consolidation, Recursive Summarization, and Semantic Drift

When an AI Agent Remembers Its Interpretation Instead of the Interaction

AI Memory and Reality Drift - Note #7 v1

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The Basic Pattern

Persistent AI agents cannot keep every interaction inside an active context window. They must select, summarize, consolidate, and retrieve past information. Detailed episodes become compact memories such as user preferences, project states, stable facts, lessons, and profiles.

This allows continuity across long periods. It also changes what persistence means. The agent may stop consulting the original interaction and begin consulting a model-generated account of what happened.

Memory becomes scalable by replacing experience with a representation of experience.

When the Summary Becomes the Memory

A first summary may preserve the main point while removing uncertainty, sequence, tone, or exceptions. Later consolidation may combine that summary with other summaries. The resulting semantic memory appears more stable and general than any individual episode.

As the history grows, the compressed memory becomes the practical past available to the agent. A provisional interpretation can harden into a user preference. A temporary state can become a permanent profile attribute. A misunderstood correction can be absorbed as another detail rather than revising the underlying belief.

How Memory Consolidation Drifts

Stage 1 - Episode. An interaction produces a detailed record with local context.

Stage 2 - Extraction. A model selects what appears important enough to preserve.

Stage 3 - Summarization. Selected details are compressed into a shorter representation.

Stage 4 - Consolidation. Multiple memories are merged into stable facts, profiles, or lessons.

Stage 5 - Recursive reuse. Consolidated memory guides later interaction and becomes input to further summaries.

Recursive Compression and Reality Drift

Recursive Compression occurs when compressed representations are repeatedly reused as source material for later compression. Information can become cleaner, shorter, and more coherent while losing contact with the events from which it originated.

Reality Drift emerges when the agent remains consistent across sessions while its remembered world becomes progressively organized around its own summaries. Continuity improves, yet the correction path back to the original interaction narrows.

The Representation Stack

Interaction becomes transcript. Transcript becomes selected memory. Selected memory becomes summary. Summary becomes profile or semantic belief. That belief shapes future responses. Future responses alter the interaction that will later be summarized.

This creates a self-confirming memory loop. A grounded memory architecture must preserve source episodes, uncertainty, time, contradiction, and the ability to revise general memories from new evidence.

Examples Across AI Systems

Personal assistants: A one-time preference is consolidated into a stable rule governing future recommendations.

Coding agents: A workaround from one environment becomes a reusable procedure applied after the conditions have changed.

Research agents: A tentative hypothesis is repeatedly summarized until it appears as an established project conclusion.

Customer agents: A disputed interaction becomes a persistent account profile that frames later support.

Conclusion

Memory consolidation is necessary for long-horizon agents. The failure lies in allowing the consolidated representation to become unanswerable to the episodes it summarizes.

A useful memory should provide continuity without manufacturing certainty. The agent must be able to remember both the conclusion and the path by which that conclusion became memory.

Related Phrases and Concepts

- agent memory
- memory consolidation
- recursive summarization
- hierarchical memory
- episodic memory
- semantic memory
- memory compression
- long-term agents

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)